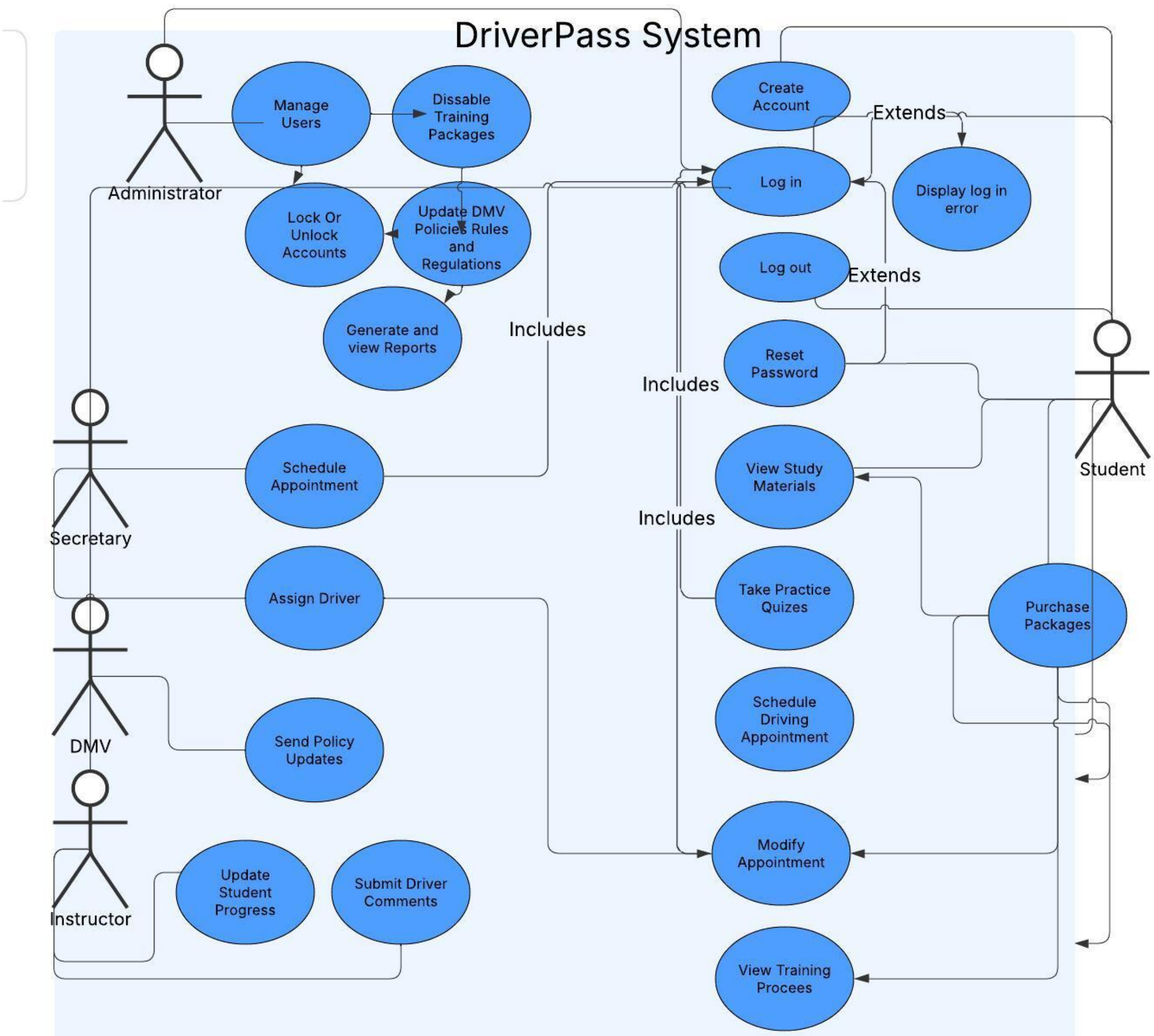
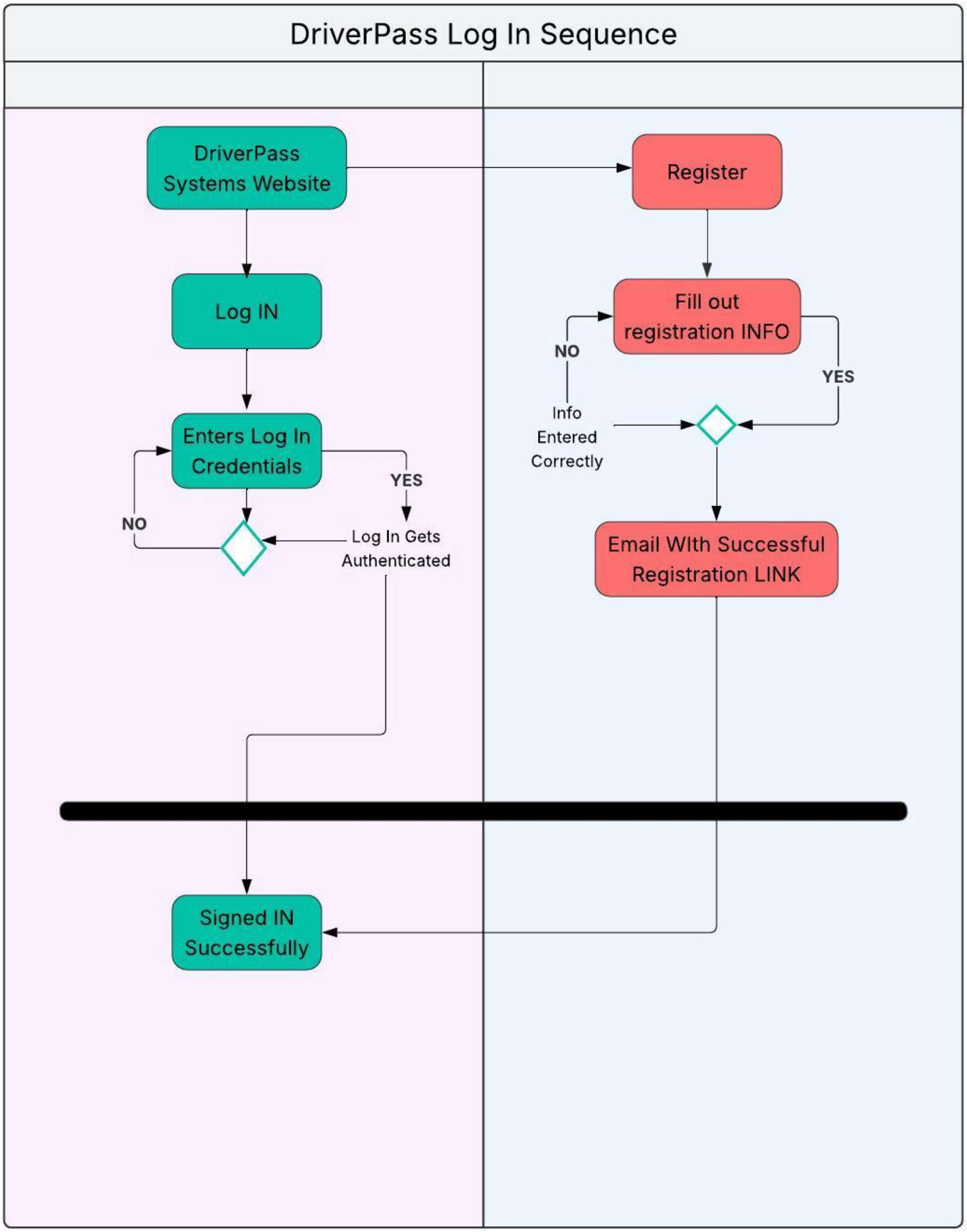
Ryan A Peguero
Cs-255

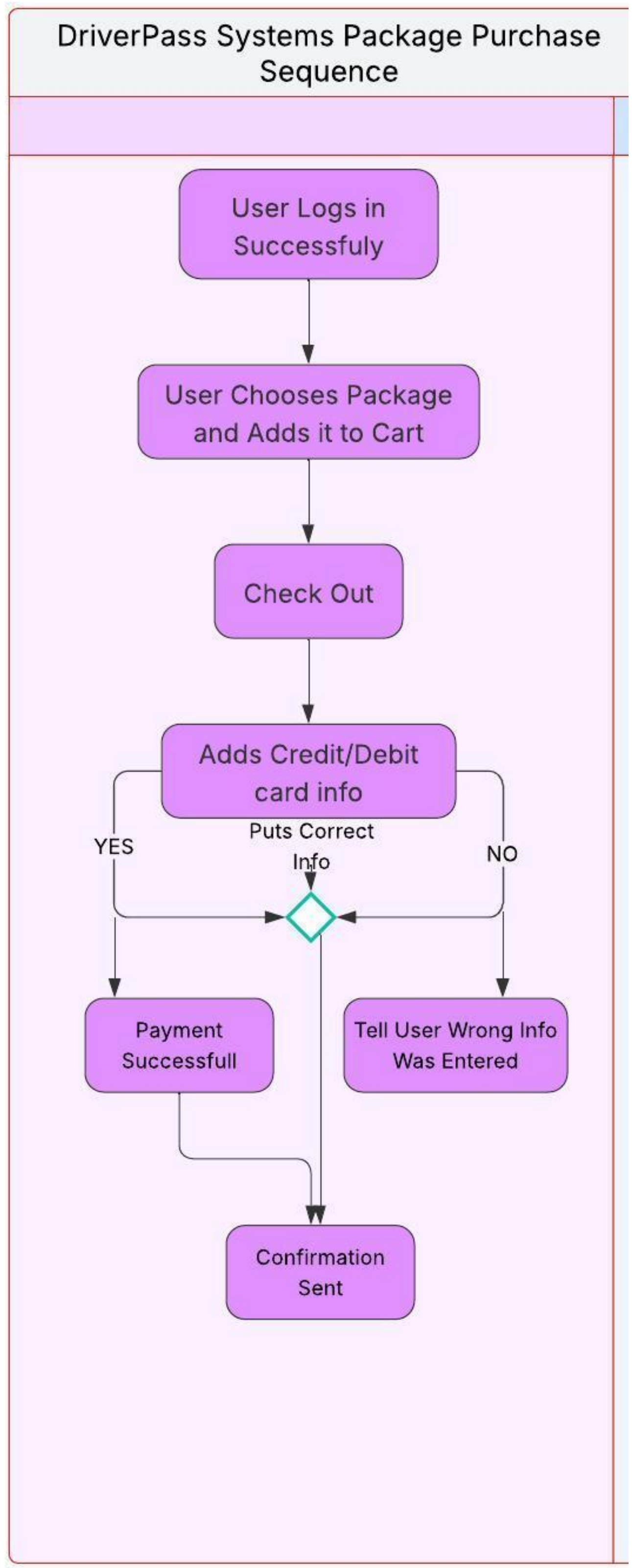
UML Diagrams

UML Use Case Diagram

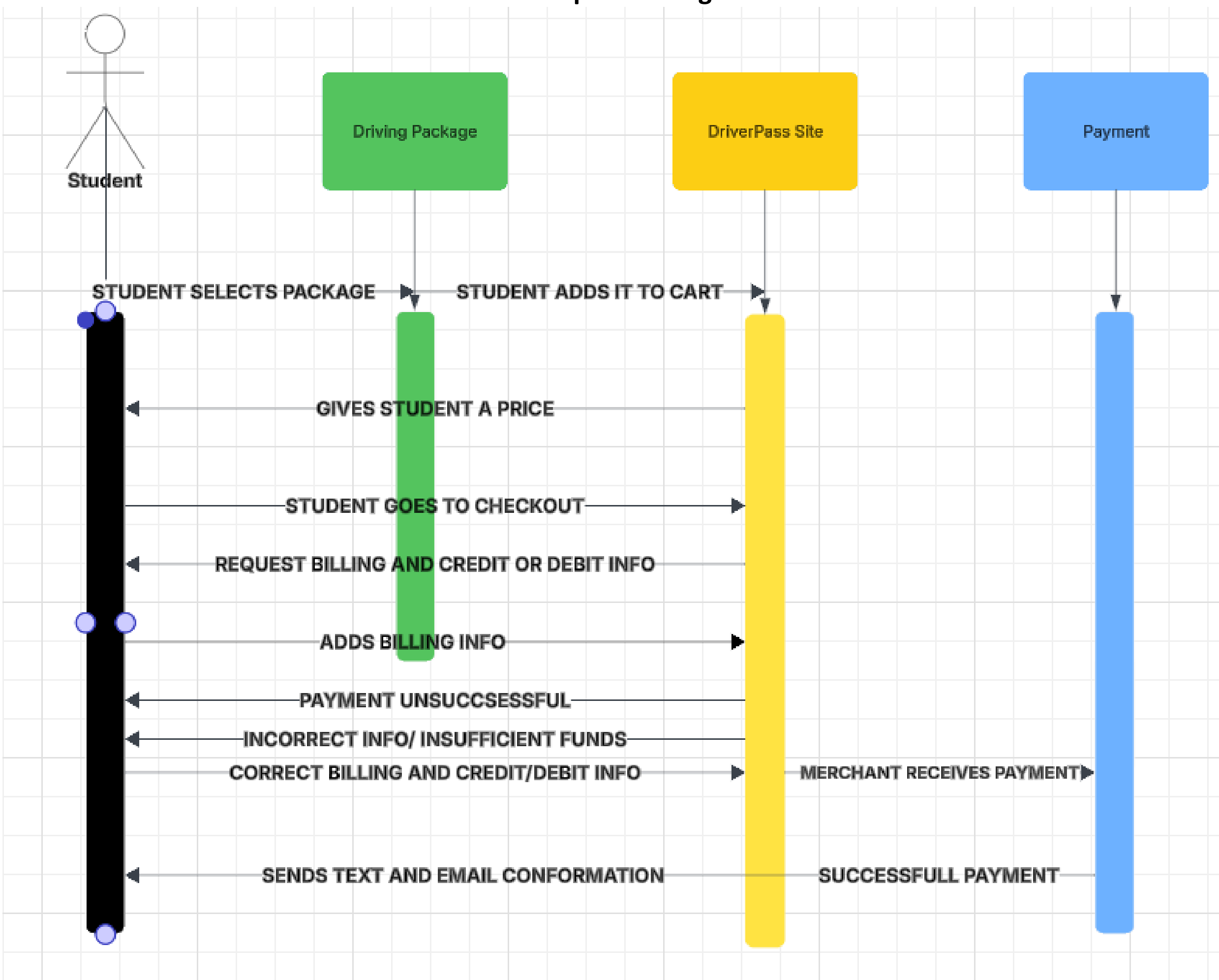


UML Activity Diagrams

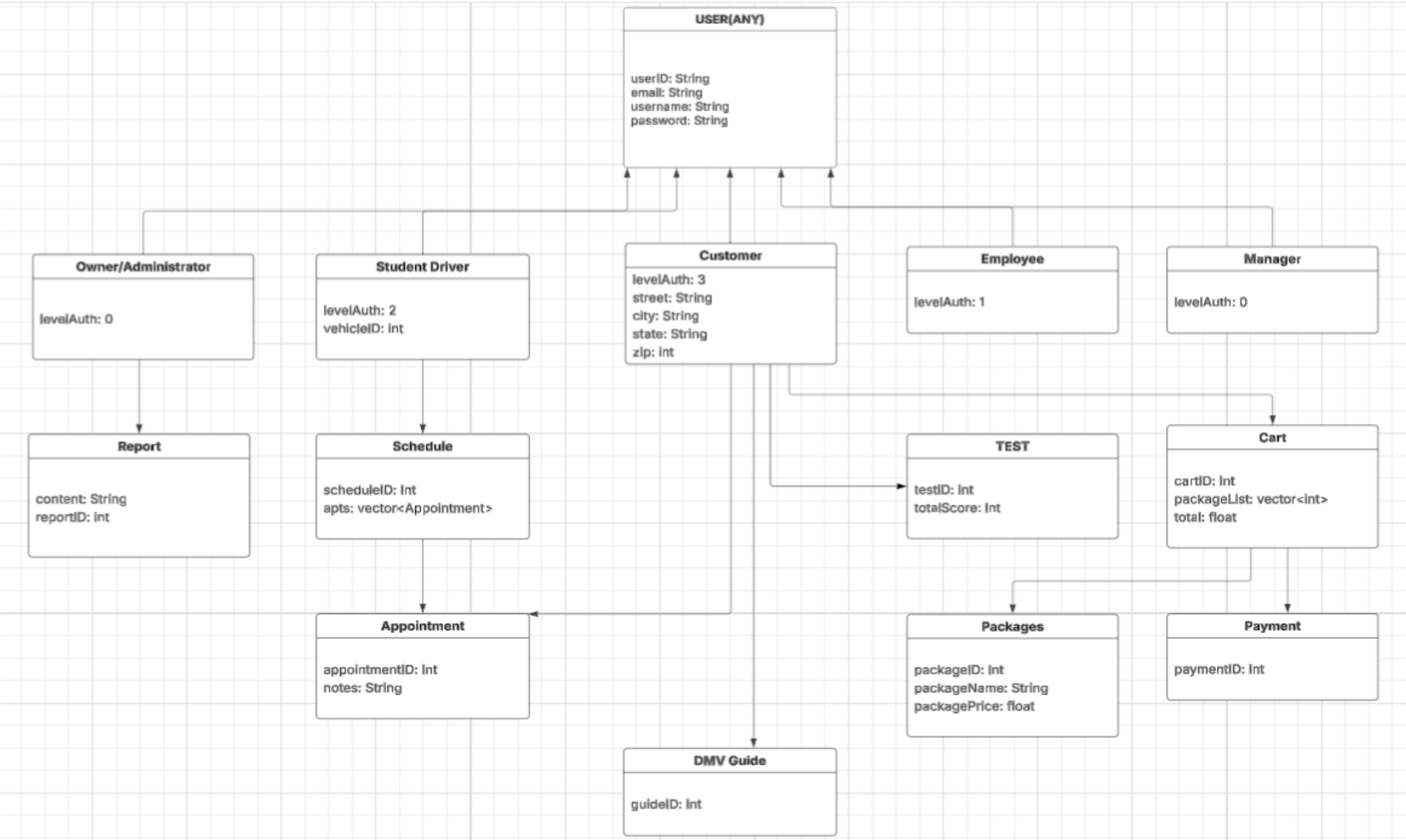



DriverPass Systems Package Purchase Sequence
User Logs in Successfuly
User Chooses Package and Adds it to Cart
Check Out
Adds Credit/Debit card info
Puts Correct Info
YES
NO
Payment Successfull
Tell User Wrong Info Was Entered
Confirmation Sent

UML Sequence Diagram



UML Class Diagram



Technical Requirements

Considering the diagrams I developed for the DriverPass platform, the system's technical framework would require a scalable, cloud-hosted environment capable of managing data related to users, including students, instructors, administrative staff, training sessions, and lesson packages. To effectively organize and maintain the relationships between these data entities, I suggest implementing a relational database management system such as MySQL due to its reliability and efficiency in handling structured data.

The application itself should be deployed as a web-based platform, utilizing JavaScript for the client-side interface to ensure interactive and responsive user experience. For the backend logic, either Python or Java would be suitable options; however, Python stands out for its simplicity and rapid development capabilities, making it my preferred choice.

In terms of development tools, Visual Studio Code would be ideal for coding due to its wide range of extensions and user-friendly interface. Git should be used for version control to manage code changes and support team collaboration effectively. Hosting the system on a cloud service like AWS or Azure will offer the flexibility needed to scale storage and processing resources as user demand increases. Additionally, implementing automated, routine backups will be essential to maintain data integrity, ensure system reliability, and support disaster recovery protocols.